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Antimicrobial mechanism of luteolin against *Staphylococcus aureus* and *Listeria monocytogenes* and its antibiofilm properties

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ABSTRACT

Luteolin (LUT) is a naturally occurring compound found in a various of plants. Few recent studies have reported LUT antimicrobial activities against bacterial pathogens, however, the fundamental LUT mediated antimicrobial mechanism has never been elucidated. This study aimed to investigate the antimicrobial activities of LUT and its mode of action against *Staphylococcus aureus* and *Listeria monocytogenes*, either as planktonic cells or as biofilms. Here, minimum inhibitory concentration (MIC), and minimum bactericidal concentration (MBC) of LUT against *S. aureus* and *L. monocytogenes* were determined using the broth microdilution method, and the antimicrobial mode of LUT was elucidated by evaluating the variations in both cell membrane integrity and cell morphology. Moreover, the biofilm inhibition was measured by crystal violet staining assay, while its qualitative imaging was achieved by confocal laser scanning microscope and field emission scanning electron microscope. MIC and MBC values of LUT against *S. aureus* were 16–32 and 32–64 μg/mL, and 32–64 and 64–128 μg/mL for *L. monocytogenes*. LUT destroyed the cell membrane integrity, as evidenced by a significant increase in the number of non-viable cells, and well-defined variations in cell morphology. Moreover, LUT presented robust inhibitory effects on the biofilm formation, enhanced antibiotics diffusion within biofilms and killed efficiently mono- and dual-species biofilm cells. Overall, LUT demonstrates potent antimicrobial properties on planktonic and biofilm cells, and the biofilm formation, and thus has the potential use as a natural food preservative in foods.

1. Introduction

Foodborne diseases remain a major public health concern worldwide and are mainly caused by the consumption of food contaminated with pathogenic organisms, including bacteria, viruses, and parasites [1]. More than 30 known foodborne pathogens have been recognized, which can cause various infectious illnesses in the human host, mostly with associated vomiting or diarrhea [2]. *Staphylococcus aureus* and *Listeria monocytogenes* represent the most common foodborne pathogens implicated in foodborne diseases. *S. aureus* is a gram-positive bacterium that causes many acute and chronic infections, and is responsible for many instances of abscesses, septicemia, arthritis, and endocarditis [3]. Therefore, extensive research has been conducted to control infections caused by *S. aureus*. Likewise, *L. monocytogenes* is a psychrophilic, gram-positive bacterium recognized as a pathogen of great importance in food, and can form biofilms and tolerate severe environmental conditions, and thus persist in foods for a prolonged period.

S. aureus and *L. monocytogenes* are able to adhere to and produce biofilms on most materials and under almost all the environmental conditions encountered in food production plants. For example, *L. monocytogenes* 10403S was shown to adhere to 17 different and food-use approved materials including metals, rubbers, and polymers [4]. Biofilm is a complex polymer composed of aggregates of microbial cells embedded in a matrix of extracellular polymeric substances, which acts as a protective "shield" against environmental stresses and antimicrobial agents [4]. Clinically, the biofilm is responsible for up to 80% of human bacterial infections, which tends to result in highly recalcitrant, chronic diseases that require dedicated therapies [5]. Similarly, in the food industry, biofilm tolerance to environmental stresses can cause the persistence resuscitation of foodborne pathogens and the recurrent cross-contamination of food products [6]. Thus, there is an urgent need to exploit novel compounds to control infections by foodborne pathogens and their biofilms [7]. Many bioactive ingredients as anti-biofilm agents have been identified from different parts of the plants [8]. Even though the last two decades have seen an explosion of studies attempting to discover compounds with a capacity for anti-

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biofilm efficacy, expanding the classes of molecules to be exploited may enhance the likelihood of exploring novel anti-biofilm agents [9].

Luteolin (LUT), which belongs to a group of substances called bioflavonoids, exerts a variety of pharmacological activities [10]. Antibacterial activities of LUT have been demonstrated against *S. aureus*, *Bacillus cereus*, *Escherichia coli*, and *Salmonella Infantis* [11], but the underlying mechanisms of LUT mediated antibacterial activities have never been reported. Therefore, the current study aimed to unravel the antibacterial activity of LUT and its mechanism of action against *S. aureus* and *L. monocytogenes*, as well as its antibiofilm activity against *S. aureus* and *L. monocytogenes* mono- and dual-species.

2. Materials and methods

2.1. Reagents

Luteolin (LUT, HPLC purity ≥ 98%) was obtained from the Chengdu Pulis Biological Science and Technology Co., Ltd, dissolved in dimethyl sulfoxide (DMSO) and filter-sterilized. All other chemicals used were of analytical grade or better.

2.2. Strains and culture conditions

Two *S. aureus* strains (*S. aureus* 1 and *S. aureus* 2) and two *L. monocytogenes* strains (*L. monocytogenes* 1 and *L. monocytogenes* 2) derived from raw goat milk in a goat farm located in Shaanxi province, China, were subjected to 16S rRNA analysis and used in the study [12]. The susceptibility profile of these strains was investigated using the VITEK® 2 compact automated system (BioMérieux®, Marcy l'Etoile, France). In addition, *S. aureus* ATCC 25923 and *L. monocytogenes* ATCC 19115 strains were employed in this study. Bacterial cells were inoculated into 3 mL of tryptic soy broth (TSB) at 37 °C with constant shaking at 200 rpm.

2.3. Determination of minimal inhibitory concentration (MIC) and minimum bactericidal concentration (MBC)

MIC and MBC assays were performed using the standard broth microdilution test as described previously [13], with slight modifications. Following twofold serial dilution of LUT in a 96 well microtiter plate (Costar, Corning, NY), and then mixed with 100 μL microbial cultures to achieve final LUT concentrations of 8–512 μg/mL and a final inoculum density of 10^6 colony forming unit (CFU)/mL. Only cultures with 1% DMSO were included as the control group. After incubation at 37 °C for 24 h, the resulting suspension was decimal diluted, inoculated onto separate tryptic soy agar (TSA) plates and incubated overnight at 37 °C for CFU counting. The MIC was determined from the lowest concentration, at which no bacterial growth occurred. The MBC value was defined as the lowest concentration of the antibacterial agent required to kill all bacterial cells.

2.4. Time-kill curves

Time-kill curves were established according to the method described by Foerster et al. [14], with minor modifications, to assess the antimicrobial effect of LUT on *S. aureus* and *L. monocytogenes* mono- and dual-species. The bacterial suspension (approximately 1 × 10^6 CFU/mL) was treated with LUT of 0, 1/4, 1/2, 1, and 2 MIC. The samples were then incubated at 37 °C at 200 rpm, and the number of viable cells was calculated every 2 h during 24 h of culture.

2.5. Biofilm viable cell counts

Biofilm viable cell was enumerated using a modified method as previously described [15]. Biofilms formed on the glass coverslip in the 24-well plate were withdrawn at predetermined time points and disrupted in a sonicating water bath. The resulting samples were rinsed three times with 200 μL of 10 mM PBS and diluted with PBS in subsequent 10-fold dilutions. Then 20 μL of the samples of each dilution were plated on TSA and incubated for 24 h at 37 °C. For each concentration at a specified time, colonies were enumerated for the dilution that produced a countable range of 30–100 colonies, and thus the CFU/mL was calculated.

2.6. Assessment of bacterial cell membrane integrity

The viability staining of bacteria cells was carried out using confocal laser scanning microscope (CLSM, Zeiss LSM 880 with Airyscan) combined with the Live/Dead BacLight Kit according to the method described by Rosenberg et al. [16], with slight modifications. The logarithmic phase cells (approximately 1 × 10^8 CFU/mL) were exposed to different concentrations of LUT (0, 1 and 2 MIC) for 4 h, centrifuged at 8000 × *g* for 6 min, and resuspended with 10 mM phosphate-buffered solution (PBS, pH 7.4). The cell suspensions were then incubated with a mixture of 2.5 μM SYTO 9 (488 nm excitation, green emission) and 15 μM propidium iodide (PI) (543 nm excitation, red emission) at 37 °C for 15 min. Subsequently, the cells were collected and washed twice with 10 mM PBS to remove excess dyes, and finally observed under a CLSM.

2.7. Cell morphology examination

Examination of the morphological changes and ultrastructural damage of the cells was performed using field emission scanning electron microscope (FESEM, Nova Nano SEM-450, FEI, Hillsboro, USA) and transmission electron microscope (TEM, G2 F20 S-TWIN, FEI, Hillsboro, USA) [17]. A volume of 20 μL of overnight cultures was added into each well of a 24-well plate and the cultures were further incubated for 3–4 h at 37 °C until approximately 10^6 CFU/mL was obtained. Then these cells were exposed to LUT at final concentrations of (0, 1, and 2 MIC) for 4 h at 37 °C. The cells were then harvested using centrifugation and prefixed with 2.5% glutaraldehyde overnight at 4 °C, and subsequently dehydrated with 25, 50, 75, 90, and 100% ethanol for 10 min in each solution. For FESEM analysis, the resulting cells were fixed on FESEM supports, sputter-coated with gold under vacuum conditions, and examined using FESEM. For TEM analysis, the resulting samples were embedded in resin, cut on ultra-microtome (MT-X, RMC, Tucson, USA), and observed using a TEM.

2.8. Biofilm analysis

Antibiofilm effects of LUT were evaluated using crystal violet staining and FESEM as described by Qian et al., with a few modifications [17]. For quantitative analysis, crystal violet staining assay was performed in a 96-well microtiter plate as previously reported [18]. Briefly, 200 μL (approximately 1 × 10^8 CFU/mL) of mono- and dual-species were exposed to varying concentrations of LUT (0, 1/32, 1/16, 1/8, 1/4, 1/2, 1, and 2 MIC), followed by a 24-h incubation at 37 °C; the biomass was measured at a wavelength of 630 nm using a microplate reader. Next, the formed biofilms were washed gently three times with 10 mM PBS to remove planktonic cells. The adherent biofilm cells were fixed with 200 μL methanol for 15 min. The plate was then dried at room temperature, and 200 μL of 0.1% crystal violet was added to each well. After incubation for 10 min, each well was washed with distilled water, and mixed with 200 μL of acetic acid (33%) to dissociate the crystal violet dye adhered to the bacterial cells. After 10 min of incubation, the absorbance was measured at 570 nm. Specific biofilm formation index was determined by calculating the ratio of biofilm measured at OD 570 over total cell growth measured at OD 630.

For qualitative analysis, 1 × 1 cm^2 coupons of glass coverslip was placed in each well of a 24-well microtiter plate. Each well was inoculated with 800 μL (approximately 1 × 10^8 CFU/mL) of *S. aureus* and

L. monocytogenes alone and in combination at a 1:1 ratio for 12 h. The resulting cultures were exposed to LUT at 0, ¼, and 1/2 MIC at 37 °C for 72 h. At three time points (24, 48, and 72 h), triplicate samples were withdrawn and analyzed. For FESEM analysis, after treatment, the coupons were kept in distilled water containing 2.5% glutaraldehyde at −4 °C for 2 h. Subsequently the coupons were washed three times with 10 mM PBS and dehydrated in graded ethanol series (30%, 50%, 90%, and 100%) for 10 min in each. Finally, the morphology of the biofilm samples was observed using FESEM.

2.9. Diffusion bioassay for gatifloxacin within biofilms

For assessment of antibiotic diffusion into the biofilm, gatifloxacin diffusion within the biofilms was determined based on its intrinsic fluorescence using CLSM [19]. The biofilms described above, which formed on glass coverslips placed inside the 24-well plate for 48 h at 37 °C in the presence of LUT at 0, 1/4, and 1/2 MIC, were then withdrawn and gently washed three times with 10 mM PBS, and a final concentration of 0.4 mg/mL gatifloxacin was added and further incubated for 4 h at 37 °C. Next, to visualize the gatifloxacin diffusion within biofilms, 2.5 μM SYTO 9 was added and incubated for 15 min. Then the samples were washed three times with 10 mM PBS and observed using a CLSM. The emission peak for gatifloxacin was recorded at 495 nm upon excitation at 291 nm. At least three random fields were visualized for each biofilm, and representative images are presented.

2.10. Evaluation of cell damages within biofilms

CLSM was employed to assess the killing effect of LUT on the dormant bacteria within the biofilm [20]. Mono and mixed species at a 1:1 ratio of *S. aureus* and L. monocytogenes were cultured for 36 h on the coupon of glass coverslip in each well of a 24-well microtiter plate, and then treated with LUT at 0, 1, 4, and 8 MIC at 37 °C for 12 h. The biofilm formed on the surface of the coupon was washed with 10 mM PBS twice, and then stained with SYTO 9 and PI. After 15 min incubation, the fluorescence intensity was measured with a microplate reader.

2.11. Statistical analysis

All experiments were repeated in triplicate, and data were analyzed using the SPSS 19.0 software (SPSS, Chicago, IL). All data are expressed as the mean values ± standard deviation (SD). Analysis of variance (ANOVA) based on the control group was carried out to determine any significant differences ($^{**}p \leq 0.01$).

3. Results

3.1. MIC and MBC of LUT against S. aureus and L. monocytogenes

According to the MIC values obtained (Table 1), *S. aureus* 1 and 2 strains investigated in this study manifested antibiotics resistance, as evidenced by MICs of penicillin (PEN, 16–32 μg/mL), erythromycin (EPY, 32–64 μg/mL), trimethoprim (TMP, 64 μg/mL), and levofloxacin (LEV, 32–64 μg/mL) (Table 1). Similarly, *L. monocytogenes* 1 and 2 strains showed resistance to trimethoprim (TMP, 2 μg/mL), imipenem (IMP, 32–64 μg/mL), levofloxacin (LEV, 128 μg/mL), and ertapenem (ETP, 32–64 μg/mL).

The results of the antibacterial activities of LUT were demonstrated in Table 1. LUT inhibited the growth of *S. aureus* and *L. monocytogenes*, and showed robust antibacterial activity against them. LUT was found to be effective with an MIC value of 16 and 32, and an MBC value of 32 and 64 μg/mL against *S. aureus* ATCC-25923 and *S. aureus* 1 and 2. *L. monocytogenes* ATCC-19115 and *L. monocytogenes* 1 and 2 were relatively resistant bacteria at MIC of 32 and 64 μg/mL, and at MBC of 64 and 128 μg/mL. Here, *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115 strains were employed for further experiments to ensure that the results were repeatable.

Table 1
MIC and MBC values of luteolin against three *Staphylococcus aureus* and three Listeria monocytogenes strains.

Strain	MIC	MBC	MIC (μg/mL)					
	Luteolin		PEN	EPY	TMP	LEV	IMP	ETP
S. aureus ATCC-25923	16	32	0.125	4	8	2	–	–
S. aureus-1	64	128	16	64	64	64	–	–
S. aureus-2	64	128	32	32	64	32	–	–
L. monocytogenes ATCC-19115	32	64	–	–	1	4	2	1
L. monocytogenes-1	128	256	–	–	2	128	64	32
L. monocytogenes-2	128	256	–	–	2	128	32	64
Dual-species	32	64	–	–	–	–	–	–

Note: PEN: Penicillin; ERY: Erythromycin; TMP: Trimethoprim; LEV: Levofloxacin; IMP: Imipenem; ETP: Ertapenem; Dual-species: the 1:1 mixture of *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115; -, Not detected.

3.2. LUT was effective against mono- and dual-species of S. aureus and L. monocytogenes

The profile of killing curves for mono- and dual-species at various concentrations of LUT was displayed in Fig. 1. LUT showed its bactericidal effect on mono- and dual-species cultures, and the rate of killing was dependent on the concentration of LUT, and was similar between strains. Mono- and dual-species were killed and seen below the limit of detection (100 CFU/mL) at MIC of LUT. Moreover, LUT at 2 MIC was more effective at killing mono- and dual-species, where *S. aureus* exposed to LUT at 2 MIC resulted in a significant decline in the number of viable cells after 24 h of treatment. In contrast, at the concentrations (1/2 and 1/4) below the MIC, cell growth occurred at a similar pace compared to that of the control.

3.3. LUT improves cell membrane permeability of S. aureus and L. monocytogenes

In principle, bacteria with intact cell membranes fluoresce bright green, whereas dead cells with compromised membranes fluoresce red. Hence, when *S. aureus* and *L. monocytogenes* cells were treated with the MIC of LUT, the live cells were enumerated using the intensity of the green fluorescence and they were markedly decreased, while the nonviable cells were enumerated using the red fluorescence and they were significantly increased as the concentration of LUT increased from MIC to 2 MIC accordingly (Fig. 2), revealing a substantial cell membrane damage within the LUT-treated cells.

3.4. LUT induced changes in cell morphology of S. aureus and L. monocytogenes

FESEM examination was performed to investigate the effect of LUT on the cell morphology of *S. aureus* and *L. monocytogenes*. As shown in Fig. 3A, untreated *S. aureus* and *L. monocytogenes* cells showed well-circumscribed contour and intact cell membrane compared to bacterial cells that were exposed to LUT. Cells treated with MIC or 2 MIC of LUT exhibited deformed shapes, which suggested a compromised membrane integrity and thus resulted in eventual cell death. Notably, morphological changes in *S. aureus* cells treated with 2 MIC of LUT became more pronounced, and similar changes were observed in the case of *L. monocytogenes* with increases in LUT concentration, where the degree of cell shrinkage increased.

Besides, as shown in the TEM images, *S. aureus* cells exposed to MIC of LUT suffered hollows and dents on their membrane surface, unlike the untreated cells with intact and well-defined membrane of the cell

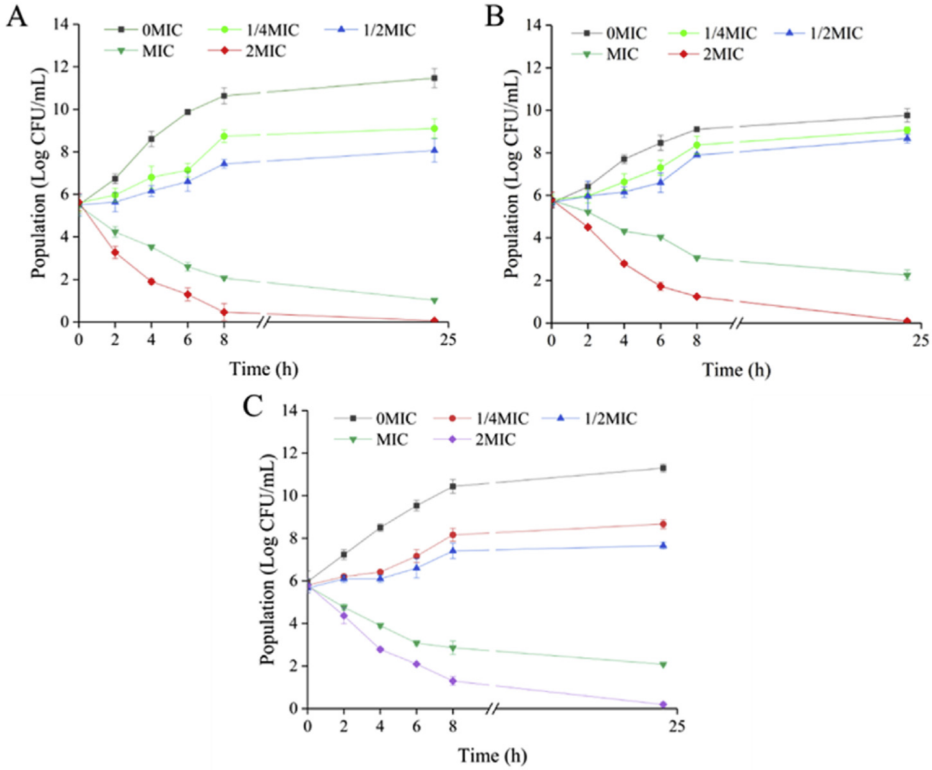

Fig. 1. Dose- and time-dependent killing curves of luteolin against *S. aureus* ATCC-25923 (A), *L. monocytogenes* ATCC-19115 (B), and dual-species (C). The number of viable cells during treatment was represented as the log reduction value of CFU counts. The error bars represent standard deviation of three replicates.

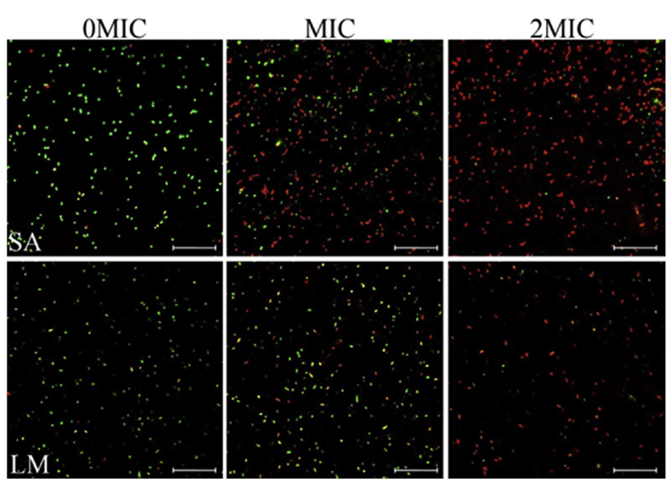

Fig. 2. Confocal laser scanning microscope observation of permeation of the bacterial cell membrane induced by luteolin. *S. aureus* ATCC-25923 (SA) and *L. monocytogenes* ATCC-19115 (LM) were incubated with luteolin at 0, 1, and 2 MIC for 4 h at 37 °C. Scale bars are 10 μm.

envelope, as observed in Fig. 3B. Similarly, *L. monocytogenes* cells treated with MIC of LUT showed partial disruptions of the outer membrane layer. In addition, with increases in LUT concentration, *S. aureus* and *L. monocytogenes* cells showed serious damage in their cell membranes and a decrease in their intracellular density, which indicated a loss in cellular integrity along with leakage of cellular components, especially for *L. monocytogenes* strains (Fig. 3B).

3.5. LUT was effective against mono- and dual-species biofilm formation in S. aureus and L. monocytogenes

As seen in Fig. 4, the biofilm formation indexes of the untreated *S. aureus* and *L. monocytogenes* mono- and dual-species were approximately 2.0, 1.6, and 2.1, respectively. For the control conditions,

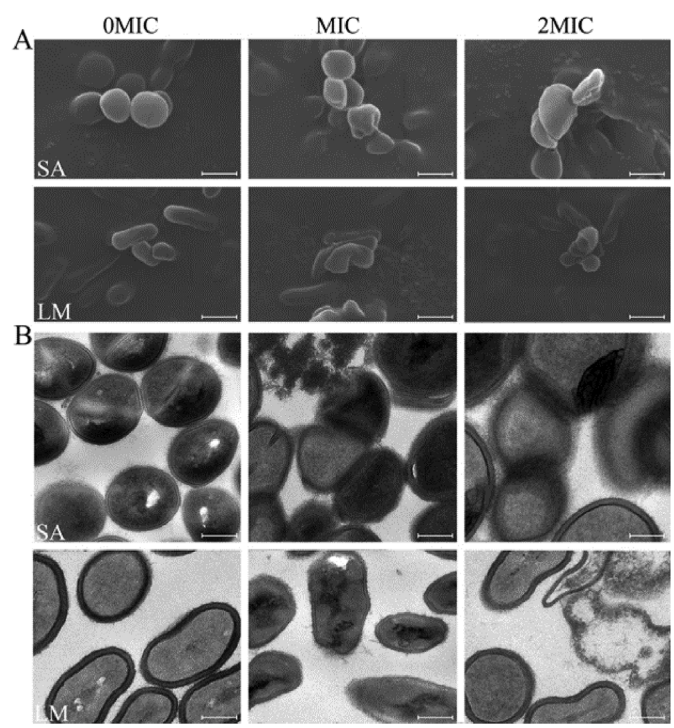

Fig. 3. Visualization of cell morphology alteration and membrane impairment using field emission scanning electron microscope (A, FESEM) and transmission electron microscope (B, TEM). The *S. aureus* ATCC-25923 (SA) and *L. monocytogenes* ATCC-19115 (LM) cells were exposed to luteolin at 0, 1, and 2 MIC for 4 h at 37 °C. Scale bars are 500 nm for FESEM and 200 nm for TEM.

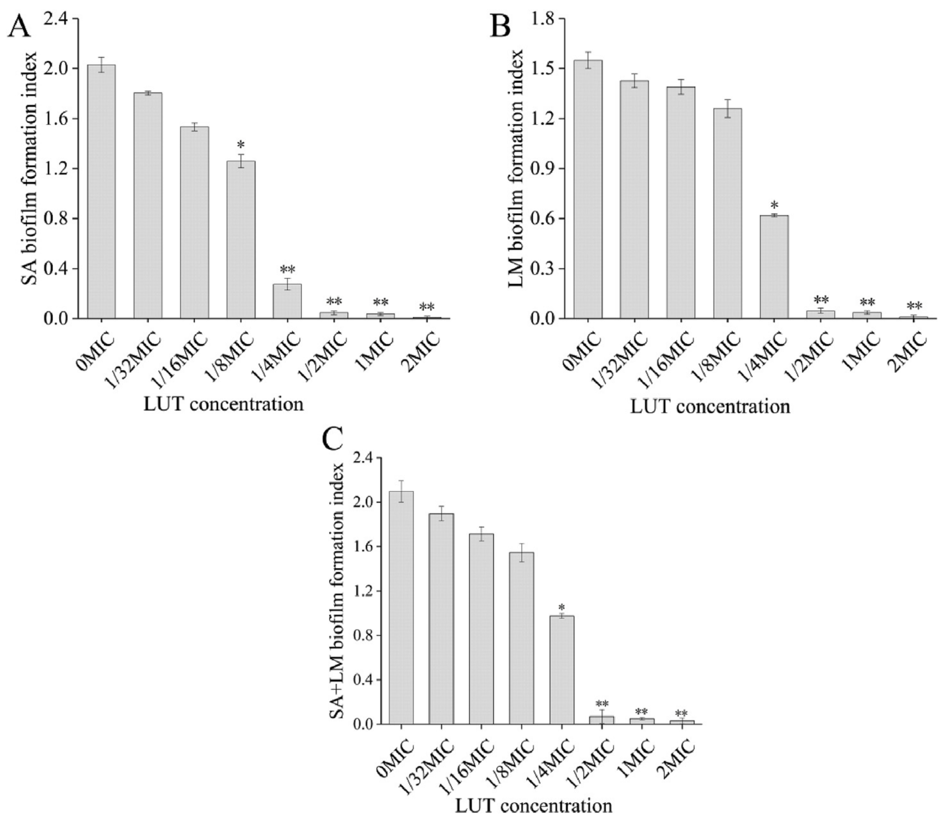

Fig. 4. Quantitative analysis of the inhibitory effect of luteolin on biofilm formation in *S. aureus* ATCC-25923 (SA, A), *L. monocytogenes* ATCC-19115 (LM, B), and dual-species (SA + LM, C) using the crystal violet staining assay. Analysis of variance (ANOVA) based on the control group was carried out to analyze any significant differences (** $p \leq 0.01$, * $p \leq 0.05$).

biofilms of dual-species had higher biofilm mass than their counterparts *S. aureus* and *L. monocytogenes* mono-species. In contrast, 1/4 MIC of LUT after 24 h of exposure showed 86.7% reduction in biofilm formation index in *S. aureus*; a similar effect was observed on *L. monocytogenes* with a reduction of 60%, but a relatively small percentage of 53.5% of biofilm inhibition was found in the case of dual-species. Hence, a higher percentage of inhibition of biofilms was observed with higher concentrations of LUT. Additionally, statistical analysis showed a negative correlation between biofilm population and LUT concentration. Following 1/2 MIC or above LUT treatment after 24 h of exposure, biofilms of both mono- and dual-species were characterized by low biovolume, and significant differences were observed between these biofilms and the control ones ($P \leq 0.05$). Thus, antibiofilm activity of LUT towards mono- and dual-species was in a concentration-dependent manner.

Next, to confirm the results obtained with the crystal violet assay, the effect of LUT on mono- and dual-species biofilm formation after 24, 48, and 72 h at 37 °C was visualized using FESEM. As shown in Fig. 5, visual inspection of biofilms demonstrated that LUT at 1/4 and 1/2 MIC could effectively inhibit biofilm formation in *S. aureus* and *L. monocytogenes* within 24 h. In the untreated group, biofilms by *S. aureus* were dense and included holes of different sizes. By contrast, when exposed to LUT at 1/4 MIC, biofilms in a monolayer manner were relatively homogenous, with occasional small clusters of bacteria while at 1/2 MIC of LUT treatment group, the number of *S. aureus* cells decreased significantly, but remained in monolayers. Similarly, in the untreated group, *L. monocytogenes* formed a biofilm consisting of dense aggregates of cells held together by extracellular polymeric substances, which were more tightly compacted with the duration of incubation. On the contrary, when exposed to LUT at 1/4 MIC, biofilms by *L. monocytogenes* consisted of cells adhered to each other, which were scattered across the surface and attached to the surface in monolayers, while in 1/2 MIC of LUT treatment group the number of *L. monocytogenes* cells declined significantly, but individual cells were observed. Moreover, in the untreated group biofilms of the dual-species, the cells were dense and composed of both species mixed together in a typical co-aggregation structure, while increasing aggregation propensity was observed throughout the incubation time. By contrast, in dual-species biofilms treated with 1/4 MIC of LUT, a close-up and top view image of 24-h biofilms presented an attached *L. monocytogenes* cell on the surface, to which grape-like clusters of *S. aureus* cells clung, with low aggregation propensity compared with that of the untreated group. Meanwhile, in 1/2 MIC of LUT treatment group, the number of dual-species cells demonstrated a pronounced decline, and amorphous and scattered low-aggregation of mono-species cells were observed compared to the mono-species biofilms. Furthermore, the low and high magnification FESEM images at 24, 48, and 72 h biofilm formation of dual-species showed that *S. aureus* grape-like aggregation increased in volume, while the ratio of *L. monocytogenes* cells was low in dual-species biofilms.

3.6. LUT treatment reduced the tolerance of S. aureus and L. monocytogenes mono- and dual-species biofilms to gatifloxacin

As shown in Fig. 6, the results displayed very little gatifloxacin diffusion in the biofilms of mono- and dual-species in the untreated group. In contrast, examination of the biofilms of mono- and dual-species demonstrated a gradual increase in gatifloxacin diffusion volume in mono- and dual-species biofilms formed in the presence of 0, 1/4, and 1/2 MIC.

3.7. LUT killed single- and dual-species biofilm cells of S. aureus and L. monocytogenes

Single- and dual-species were found to form biofilms, which were largely stained with SYTO 9, and damaged or dead cells within the biofilm were labelled with PI. The biofilms of mono- and dual-species formed on the surface of a glass coverslip harbored 10^7–10^8 CFU/coverslip of viable bacteria. As shown in Fig. 7, visual inspection of LIVE/DEAD stained wells showed that treatment of preformed mono- and dual-species biofilms with increasing concentrations of LUT for 12 h

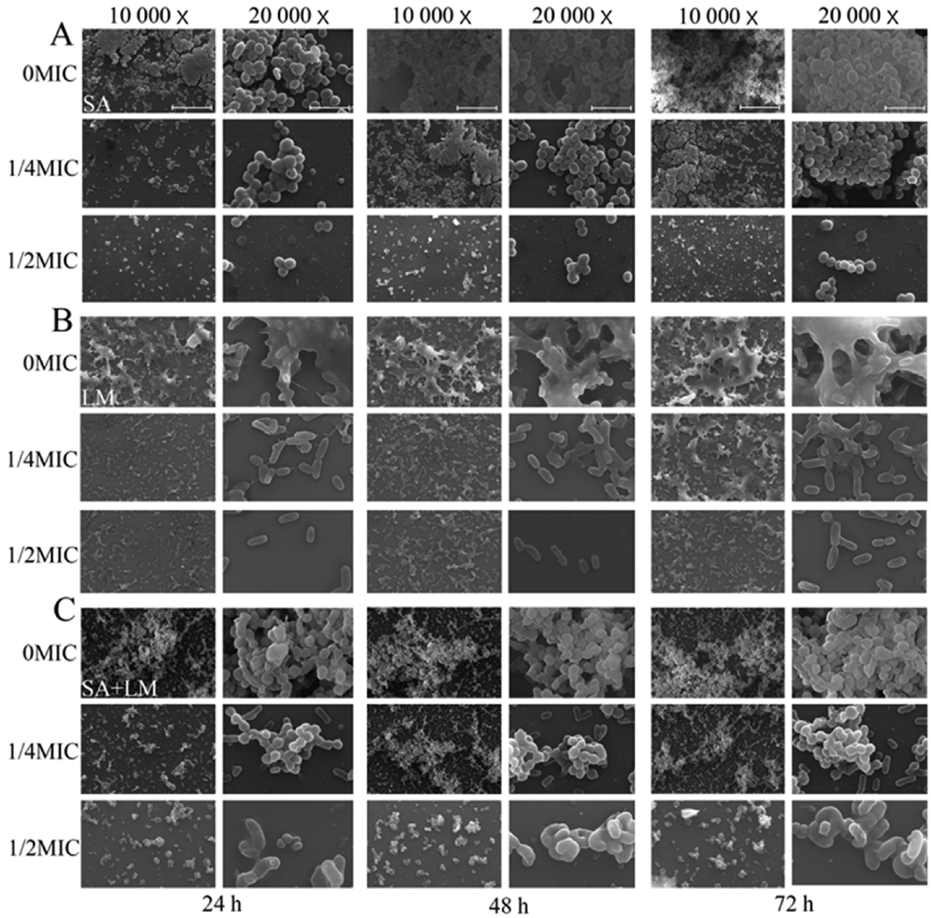

Fig. 5. Qualitative analysis of the effect of luteolin on the biofilm formation of *S. aureus* ATCC-25923 (SA, A), *L. monocytogenes* ATCC-19115 (LM, B) and dual-species (SA + LM, C) using a field emission scanning electron microscope (FESEM). Representative FESEM images (low and high magnification, × 10 000 and 20 000) of mono- and dual-species were observed. Scale bars are 5 and 1 μm, respectively.

presented a dose-dependent killing effect on bacterial cells embedded in biofilms, where treatments with higher concentrations of LUT resulted in the progression from viable fluorescence to nonviable fluorescence with increasing scope and depth in the biofilm.

From 0 to 2 MIC, the viability profile in both *L. monocytogenes* mono- and dual-species biofilms changed a little, whereas at 2 MIC of LUT, the nonviable fluorescence in *S. aureus* mono-species biofilms was observed, and it shifted deeper into the image stack. Upon exposure to 4 MIC of LUT, in both *L. monocytogenes* mono- and dual-species biofilms, nonviable fluorescence was shown to be markedly increased, while it required a slightly higher concentration of LUT (8 MIC) to obtain a similar killing effect on *L. monocytogenes* mono-species biofilm cells. Under similar conditions, treatment of dual-species biofilms with 8 MIC of LUT led to a low reduction in the number of viable cells compared to those of *L. monocytogenes*. Moreover, although almost all of the cells within the biofilm of single- and dual-species emitted red fluorescence when treated with LUT at 8 MIC, there were a few locations located deep inside the biofilm clusters that produced green fluorescence, indicating that several cells within these biofilms were still intact and viable.

4. Discussion

The increasing microbial resistance to classical antibiotics has stimulated the exploitation of potential compounds to combat bacterial infections. In this sense, the current study evaluated LUT antibacterial and antibiofilm effects on two important foodborne pathogens, aiming to co-potentiate the antimicrobial effect of the most commonly used preservatives in the food industry. Concerning the antibacterial activity, MIC values of LUT were 16 and 32 μg/mL for *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115, respectively. This is lower than a previous report where MIC of grape pomace extract against *S. aureus* ATCC-8275 was 500 μg/mL [21]. Also, González-Alamilla et al. reported the antibacterial activity of *Salix Babylonica* L. hydroalcoholic extract against multi-resistant *S. aureus* and L. monocytogenes with MICs of 390 and 780 μg/mL, respectively [22].

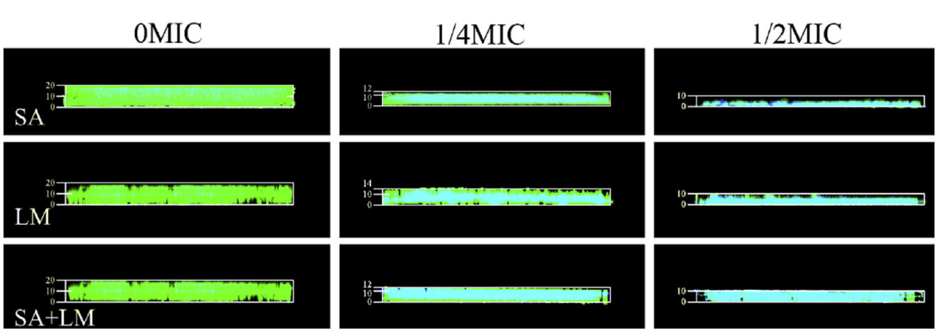

Fig. 6. Representative CLSM images assessing the diffusion of gatifloxacin through biofilms formed in the presence of different concentrations of luteolin. The biofilms of *S. aureus* ATCC-25923 (SA), *L. monocytogenes* ATCC-19115 (LM) and dual-species (SA + LM) were stained with SYTO 9 for biofilms (green) and the intrinsic fluorescence of gatifloxacin (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

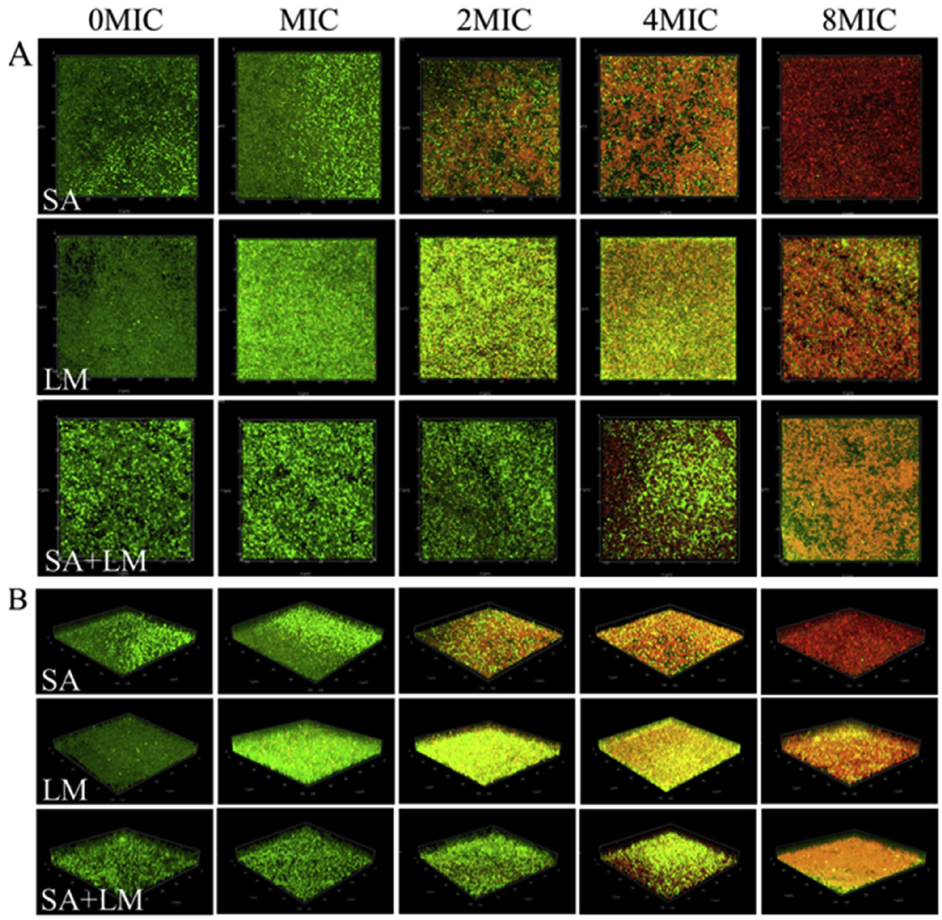

Fig. 7. The killing effect of luteolin against *S. aureus* ATCC-25923 (SA), *L. monocytogenes* ATCC-19115 (LM), and dual-species (SA + LM) cells within biofilms using a confocal laser scanning microscope (CLSM). Representative CLSM plane (A) and three-dimensional (B) photographs are shown.

Furthermore, BacLight LIVE/DEAD stain has been broadly used as an indicator of cell viability. In our study, CLSM images showed that LUT was capable of destroying the cell membrane of *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115, illustrated by the accumulation of enormous number of red-fluorescent bacterial cells, whereas untreated cells remained intact, as evidenced by the presence of many green-fluorescent bacteria cells. Similar phenomena were also observed in LUT-treated cells of *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115 using FESEM and TEM studies. Following LUT treatment, both *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115 cells appeared deformed, with destroyed cell membrane as seen using FESEM. Additionally, TEM results revealed that *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115 exposed to LUT showed partial or complete destruction of the cell membrane and leakage of the intracellular contents compared with the control group. Kim et al. reported that TEM images of *E. coli* O157:H7 ATCC-35150, *Salm. enteritidis* USDA-FSIS 15060, and *L. monocytogenes* ATCC-19115 treated with 5.0% arrowroot tea revealed the rupture of cell walls and non-homogeneous disposition of cytoplasmic materials within the treated bacteria [23]. A similar report also suggested that a significant cell membrane damage and leakage of the intracellular contents of gram-positive and gram-negative bacteria treated with graphene oxide were observed using FESEM and TEM [24].

Biofilms are medically critical because they have been implicated in approximately 80% of chronic and recurrent microbial infections in the human body, which are often very difficult to treat successfully due to their intrinsic resistance to antibiotics [25]. Here, LUT at 1/2 MIC or 1/4 MIC was able to significantly reduce biofilm formation in *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115 mono- and dual-species. These results are consistent with a previous observation by Chen et al. that baicalein inhibited biofilm formation in *S. aureus* 17546 [26]. Similarly, Du et al. reported that sub-MIC and MIC of epigallocatechin-gallate could significantly reduce biofilm formation in *L. monocytogenes* ATCC-19114 on polystyrene microtiter plates [27]. In addition, the three-dimensional structure of the biofilm by *L. monocytogenes* cells was different in mono- and dual-species biofilms, but *L. monocytogenes* cells were mainly attached to the surface in the presence and absence of *S. aureus*. A similar study reported that scanning electron microscope revealed higher sessile populations of *L. monocytogenes* EGD-e, and an intimate association between *L. monocytogenes* EGD-e and *S. aureus* CIP 53.156 was observed in dual-species biofilms [28]. Specifically, LUT treatment markedly enhanced the diffusion of gatifloxacin within biofilms of *S. aureus* and *L. monocytogenes* mono- and dual-species in the presence of 1/2 or 1/4 MIC LUT, which indicated that LUT increased the permeability of biofilms of *S. aureus* and *L. monocytogenes* mono- and dual-species. The results demonstrated that LUT treatment can potentially help to kill robust biofilm-based bacteria using antibiotics.

LUT also exhibited the capability of killing cells within *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115 mono- and dual-species biofilms. CLSM images displayed that *S. aureus* ATCC-25923 and *L. monocytogenes* ATCC-19115 cells within biofilms of single culture for both strains were almost red when treated with 8 MIC of LUT, revealing that nearly all biofilm cell membranes were impaired. A previous study suggested that phenylacetic acid was effective in inactivating *L. monocytogenes* cells embedded in mature biofilms [15]. Moreover, most of these cells within mixed-species biofilms appeared red when treated with 8 MIC of LUT, indicating that LUT remained active against cells within the mixed-species biofilms though they were much harder to kill than those within biofilms in pure culture.

5. Conclusions

To our knowledge, this is the first study assessing the antimicrobial effect and mechanism of action of LUT against *S. aureus* and *L. monocytogenes* in both planktonic and biofilm states. The results of the present study clearly show that LUT exerts its potent antibacterial effect by impairing bacterial cell membranes and inducing cell morphological alterations in the planktonic state, as well as antibiofilm activities, by inhibiting biofilm formation and killing biofilm cells. These findings suggest that LUT may have valuable applications in the food industry as a food preservative and surface disinfectant.

CRediT authorship contribution statement

Weidong Qian: Conceptualization, Methodology, Supervision. **Miao Liu:** Software, Validation, Writing - original draft. **Yuting Fu:** Formal analysis, Writing - original draft, Resources. **Jianing Zhang:** Methodology, Investigation, Data curation. **Wanting Liu:** Writing - review & editing, Visualization. **Jingyuan Li:** Methodology, Investigation. **Xiang Li:** Writing - review & editing. **Yongdong Li:** Resources, Formal analysis. **Ting Wang:** Writing - review & editing, Supervision, Methodology.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.micpath.2020.104056.

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